
IGGS-Merge: Information-Geometric Gradient Surgery for Robust Test-Time Model Merging

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Abstract

Test-Time Model Merging (TTMM) dynamically solves for layer-wise merging coefficients to combine specialized expert models on unlabeled test streams, offering a backpropagation-free alternative to traditional test-time adaptation. However, existing TTMM methods either optimize merging coefficients on a flat Euclidean space that ignores parameter-level sensitivities, or suffer from severe gradient interference under environmental corruptions. To address these fundamental limitations, we propose IGGS-Merge (Information-Geometric Gradient Surgery), a mathematically rigorous and robust test-time model merging framework. IGGS-Merge reformulates test-time gradient-based updates in a Riemannian space where the metric tensor is defined by the pre-computed parameter-level joint diagonal Fisher Information of the experts. Specifically, we project conflicting class-specific gradients onto each other’s normal planes in this Riemannian space using a Fisher-weighted inner product and norm, while adaptively damping updates in highly sensitive layers and accelerating them in robust representational layers. Exhaustive evaluations on a multi-task vision stream (MNIST, FashionMNIST, KMNIST) under severe environmental corruptions demonstrate that IGGS-Merge out-performs all existing state-of-the-art methods, establishing a new state-of-the-art for TTMM.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

Test-Time Adaptation (TTA) has emerged as a crucial paradigm for deploying deep learning models under non-stationary environmental shifts and continuous out-of-distribution (OOD) test streams (Wang et al., 2021a). Traditional TTA methods rely on performing backpropagation and parameter updates on the full model at inference time. While effective, this process is computationally expensive, highly sensitive to learning rates, and prone to catastrophic forgetting or representation collapse under continuous task shifts.

To overcome these computational bottlenecks, Test-Time Model Merging (TTMM) has recently been proposed as a lightweight, backpropagation-free alternative (Yang et al., 2024b; Authors, 2026c). TTMM operates on a library of pre-trained, specialized “expert” models (or adapters like LoRAs) that lie in the same loss basin. Instead of adapting millions of model parameters at test time, TTMM dynamically solves for a set of low-dimensional merging coefficients (e.g., layer-wise or global task-routing weights) to combine the expert parameters linearly on-the-fly, creating a unified model tailored to the current batch’s distribution.

However, existing teacher-free TTMM methods suffer from two critical limitations:

1. Catastrophic parameter drift: Under sequential streams (long blocks of homogeneous tasks), prediction entropy minimization drives the merging coefficients entirely towards the active task, causing decision-boundary collapse. When the task shifts, the model is permanently trapped in a sub-optimal state and fails to adapt.
2. Gradient interference: Under severe environmental corruptions (noise, blur, or contrast shift), the predictions become noisy, and gradients from different classes in the same batch pull the low-dimensional merging coefficients in opposite directions, causing severe optimization instability and representation collapse.

To break these fundamental limits, recent works have

proposed Optimizer and Parameter Resets (OPR) to detect task boundaries and reset parameters to uniform (Authors, 2026b), and class-specific gradient surgery (PC-Merge) to project conflicting class gradients. While effective, standard gradient surgery performs projections in a flat Euclidean coefficient space. This ignores a fundamental property of deep neural networks: different layers exhibit highly heterogeneous representational and structural sensitivity. Specifically, classification-adjacent layers and early convolutional layers are extremely sensitive to parameter updates, whereas intermediate representational layers are highly robust (Authors, 2026a).

In this paper, we propose IGGS-Merge (Information-Geometric Gradient Surgery), a mathematically principled framework that unifies information geometry with multi-task gradient surgery. Our core insight is that the coefficient space has a non-trivial information geometry induced by the parameter-space diagonal Fisher Information.

Our contributions are summarized as follows:

- We propose IGGS-Merge, which reformulates test-time gradient surgery in a Riemannian coefficient space where the metric tensor is defined by the joint diagonal Fisher Information of the experts.
- We mathematically derive and prove the Riemannian orthogonality of our projection scheme, showing why standard Euclidean surgery fails to eliminate gradient interference under heterogeneous sensitivities.
- We present a complete, detailed, step-by-step algorithm box for our Information-Geometric Gradient Surgery, integrating online preconditioned updates with optimizer resets.
- Through exhaustive evaluations on MNIST, FashionMNIST, and KMNIST streams under severe environmental corruptions, we show that IGGS-Merge significantly outperforms all existing baselines, achieving state-of-the-art results.
- We present an empirical breakdown of structural sensitivities in ResNet-18, a thorough ablation study, and a sensitivity analysis of our damping factor α , providing valuable guidance for robust test-time adaptation.

2. Related Work

2.1. Test-Time Adaptation

Test-Time Adaptation (TTA) aims to adapt pre-trained neural networks to out-of-distribution (OOD) test streams on-the-fly without access to the training data. The pioneering work Tent (Wang et al., 2021a;b) minimizes prediction entropy to update affine parameters in batch normalization layers. While simple and effective, standard TTA suffers from severe instability under continuous and non-stationary task shifts. To address this, various methods have been proposed to stabilize TTA, such as using contrastive loss (Chen et al., 2022), Bayesian estimation (Zhou et al., 2025), or handling complex multimodal noises (Guo & Jin, 2025). In continual scenarios, models are prone to catastrophic forgetting and representation collapse. Researchers have developed memory-efficient adaptation (Song et al., 2023), prompt-based continuous adaptation (Gan et al., 2022), source-free stable updates (Arasu & Jahangiri, 2025), class-wise fine-tuning (Zhao et al., 2025), and correlation-robust methods (Gong et al., 2022). More recently, Lee et al. (Lee et al., 2024b;a) and Han et al. (Han et al., 2025) analyzed the shortcomings of pure entropy minimization and proposed improved objective functions. A major challenge is addressing pathological TTA under extreme shifts, where simple baselines like R-DUMB (Press et al., 2023; Mishra, 2026) can sometimes outperform complex adaptive methods. Additionally, the role of batch normalization statistics has been heavily scrutinized in continual environments (Lange et al., 2023; Shao et al., 2025). While these methods advance the field, they require backpropagation through the model’s large parameter space, incurring substantial computational costs and being prone to divergence.

2.2. Model Merging and Task Vectors

To bypass expensive test-time backpropagation, Test-Time Model Merging (TTMM) has emerged as a promising alternative (Yang et al., 2024b; Authors, 2026c). Model merging combines independent specialized expert models fine-tuned from the same base network to resolve multi-task trade-offs without further training. Early weight averaging approaches such as Stochastic Weight Averaging (SWA) (Izmailov et al., 2018) or Mean Teacher (Tarvainen & Valpola, 2017a;b) averaged parameters for stability. Modern model merging often utilizes task vectors (Saadati et al., 2024), which are constructed by subtracting base parameters from fine-tuned weights, allowing model editing, multi-task learning, and deep representation surgery (Yang et al., 2024a). Fisher-weighted averag-

ing (Matena & Raffel, 2021) pioneered the use of diagonal Fisher Information to merge parameters based on sensitivity. Recent works like AdaMerging (Yang et al., 2024b; 2023) learn optimal task-routing merging coefficients adaptively. Advanced optimization algorithms such as Frank-Wolfe optimization (Chen et al., 2025) and unified generalization frameworks (Li et al., 2026; Khan et al., 2024) have been developed to scale model merging to large-scale vision-language models (Karamanov et al., 2024) and multi-task policies (Lawson & Qureshi, 2023). Curvature-aware and layer-adaptive merging techniques such as CAMEX (Dung et al., 2025) and data-free layer-adaptive merging via Fisher Information (Xia, 2026) have further demonstrated the importance of considering non-trivial curvature in parameter space. S2C-Merge (Authors, 2026c), PC-Merge (Authors, 2026b), and LFWA (Authors, 2026a) bring model merging into the online streaming setting, but still suffer from representation collapse or gradient conflicts in heterogeneous spaces.

2.3. Multi-Task Gradient Surgery

In multi-task learning and test-time adaptation, gradients computed from different objectives or classes often conflict, leading to optimization instability. Gradient Surgery (PCGrad) (Yu et al., 2020) projects the gradient of one task onto the normal plane of another if their inner product is negative. This technique has been widely applied in domain generalization (Manzilla et al., 2021), multitask finetuning (CS224N et al., 2023; Noneb), and orthogonal task projections (Yadav et al., 2025). Standard gradient surgery, however, assumes a flat Euclidean parameter space. In heterogeneous deep architectures, standard Euclidean projections are insufficient and fail to guarantee orthogonality in the true Riemannian manifold of the network, which we resolve in this paper.

3. Methodology

3.1. Problem Formulation

Let θ_{base} be the parameters of a pre-trained base model. We have K specialized expert models finetuned from the same checkpoint on different tasks $\{T_1, \dots, T_K\}$. For each expert k , we define its task vector as $v_k = \theta_k - \theta_{\text{base}}$. We define layer-wise merging coefficients $\Lambda = [\lambda_1, \dots, \lambda_K]^T$ constrained to the unit simplex:

$$\sum_{k=1}^K \lambda_{k,w} = 1, \quad \lambda_{k,w} \geq 0 \quad \forall w \in \mathcal{L} \quad (1)$$

where $\mathcal{L}$ is the set of named parameter tensors in the backbone. The virtual merged model for tensor w is:

$$w_{\text{merged}}(\Lambda) = w_{\text{base}} + \sum_{k=1}^K \lambda_{k,w} v_{k,w} \quad (2)$$

During test-time adaptation, the model receives a continuous stream of unlabelled batches X_t . The goal is to dynamically solve for the simplex-constrained coefficients Λ at each online step t using X_t .

3.2. Riemannian Geometry of the Coefficient Manifold

The space of merging coefficients Λ forms a low-dimensional optimization manifold embedded within the model’s massive parameter space. A fundamental property of deep neural networks is that different parameters exhibit vastly different sensitivities: a small perturbation in early convolutional features or final classifier weights can catastrophically alter the prediction distribution, while intermediate residual blocks remain highly robust.

To formalize this structural sensitivity on our coefficient manifold, we define a Riemannian metric tensor G on the coefficient space. For each parameter tensor w , the metric G_w is defined using the joint diagonal Fisher Information F_w :

$$G_w = (F_w + \epsilon_{\text{scale}})^\alpha \quad (3)$$

where $\alpha \geq 0$ is a sensitivity damping hyperparameter, and $\epsilon_{\text{scale}} > 0$ is a small scale stabilization constant. The joint diagonal Fisher Information F_w is pre-computed on a small, clean calibration dataset of the experts’ training distributions:

$$F_w = \frac{1}{K} \sum_{k=1}^K F_w^{(k)} \quad (4)$$

where $F_w^{(k)}$ is the diagonal Fisher Information of expert k for tensor w , computed as:

$$F_w^{(k)} = \frac{1}{|D_{\text{cal}}|} \sum_{(x,y) \in D_{\text{cal}}} (\nabla_w \log p(y|x; \theta_k))^2 \quad (5)$$

The joint Fisher Information F_w serves as a local diagonal quadratic approximation of the Kullback-Leibler (KL) divergence in the model’s output distribution space. Under this Riemannian metric, the distance between two nearby coefficient configurations Λ and $\Lambda + d\Lambda$ is given by:

$$ds^2 = \sum_{w \in \mathcal{L}} G_w \|d\Lambda_w\|_2^2 \quad (6)$$

This formulation ensures that changes in coefficients of highly sensitive layers correspond to a larger Riemannian distance, while updates in robust representational layers are mathematically shrunk, matching the true geometry of the model’s prediction manifold.

3.3. Information-Geometric Gradient Surgery (IGGS-Merge)

During test-time adaptation, the model receives an unlabeled batch of data X_t . Let C be the number of classes. We perform a forward pass on the batch X_t to obtain predicted class pseudo-labels. For each class c present in the batch, we isolate the subset of predictions and backpropagate class-specific objectives (such as class-specific entropy) to compute class-specific gradients with respect to the merging coefficients:

$$g_c = \nabla_{\Lambda} \mathcal{L}_c(X_t; \Lambda) \quad (7)$$

where $\mathcal{L}_c$ is the entropy loss computed over the subset of the batch predicted as class c .

In flat Euclidean space, standard gradient surgery (PC-Merge) resolves conflicts by projecting conflicting gradients. However, because it assumes a flat identity metric I , it treats all layers as equally sensitive, resulting in catastrophic interference in highly sensitive layers.

To resolve this, we propose performing gradient surgery directly in the Fisher Riemannian tangent space. For two class-specific gradients g_a and g_b , their information-geometric inner product is defined as:

$$\langle g_a, g_b \rangle_F = \sum_{w \in \mathcal{L}} G_w \cdot (g_{a,w} \cdot g_{b,w}) \quad (8)$$

and the squared Fisher norm is defined as:

$$\|g_b\|_F^2 = \langle g_b, g_b \rangle_F \quad (9)$$

If $\langle g_a, g_b \rangle_F < 0$, indicating an information-geometric conflict between the updates for class a and class b , we project g_a onto the normal plane of g_b in the Riemannian manifold:

$$g_a^{\text{projected}} = g_a - \frac{\langle g_a, g_b \rangle_F}{\|g_b\|_F^2 + \epsilon} g_b \quad (10)$$

where $\epsilon > 0$ is a small numerical safety parameter.

The final, conflict-free gradient is the sum of these projected class gradients:

$$g_{\text{final}} = \sum_c g_c^{\text{projected}} \quad (11)$$

Algorithm 1 Information-Geometric Gradient Surgery (IGGS-Merge)

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1: Input: Base model parameters  $\theta_{\text{base}}$ , expert models  $\{\theta_k\}_{k=1}^K$ , pre-computed diagonal Fisher Information matrices  $\{F^{(k)}\}_{k=1}^K$ , learning rate  $\eta$ , damping factor  $\alpha$ , stability constant  $\epsilon_{\text{scale}}$ , reset threshold  $\tau$ .
2: Compute Joint Fisher:  $F_w = \frac{1}{K} \sum_k F_w^{(k)}$  and metric  $G_w = (F_w + \epsilon_{\text{scale}})^{\alpha}$  for all tensors  $w \in \mathcal{L}$ .
3: Initialize: Merging coefficients  $\Lambda_0 = [1/K, \dots, 1/K]^T \in \mathbb{R}^{K \times |\mathcal{L}|}$ .
4: for each online step  $t = 1, 2, \dots$  do
5:   Receive unlabeled batch  $X_t$ .
6:   Construct merged model:  $w_t = w_{\text{base}} + \sum_k \lambda_{k,w,t} v_{k,w}$ .
7:   Forward pass on  $X_t$  to obtain predictions and pseudo-labels  $\hat{y} = \arg \max P(y|X_t; \theta_t)$ .
8:   if Loss spike detected (OPR reset condition) then
9:     Reset coefficients to uniform:  $\lambda_{k,w,t} = 1/K$ .
10:    Re-merge model  $w_t$ .
11:   end if
12:   Group batch into class-specific subsets  $X_t^c = \{x \in X_t \mid \hat{y} = c\}$ .
13:   for each class  $c$  present in batch do
14:     Compute class-specific entropy loss:  $\mathcal{L}_c = -\frac{1}{|X_t^c|} \sum_{x \in X_t^c} \sum_y P(y|x) \log P(y|x)$ .
15:     Compute class gradient:  $g_c = \nabla_{\Lambda} \mathcal{L}_c$ .
16:   end for
17:   Gradient Surgery (Riemannian Projection):
18:   Initialize projected gradients  $g_c^{\text{projected}} = g_c$  for all  $c$ .
19:   for each pair of classes  $a, b$  present in batch do
20:     Compute Fisher-weighted inner product:  $\langle g_a, g_b \rangle_F = \sum_w G_w (g_{a,w} \cdot g_{b,w})$ .
21:     if  $\langle g_a, g_b \rangle_F < 0$  then
22:       Project:  $g_a^{\text{projected}} \leftarrow g_a^{\text{projected}} - \frac{\langle g_a, g_b \rangle_F}{\|g_b\|_F^2 + \epsilon} g_b$ .
23:     end if
24:   end for
25:   Aggregate and Update:
26:    $g_{\text{final}} = \sum_c g_c^{\text{projected}}$ .
27:   Preconditioned update:  $\lambda_{k,w,t+1} = \lambda_{k,w,t} - \frac{\eta}{G_w} g_{\text{final},w}$ .
28:   Simplex projection:  $\Lambda_{t+1} = \Pi_{\Delta}(\Lambda_{t+1})$ .
29: end for

```

We then perform a preconditioned update step on the merging coefficients, matching the natural gradient descent step on the Riemannian manifold:

$$\lambda_{k,w} \leftarrow \lambda_{k,w} - \frac{\eta}{G_w} g_{\text{final},w} \quad (12)$$

where $\eta > 0$ is the learning rate. This preconditioning guarantees that sensitive layers (where G_w is large) undergo small, controlled adaptations, preventing representation collapse, while robust layers adapt rapidly to capture task-specific features. After the update, the coefficients are projected back onto the unit simplex via simplex projection Π_{Δ} to satisfy the probability constraint.

3.4. Theoretical Analysis and Proofs

We now formalize the mathematical rigor of our proposed projection scheme.

Proposition 3.1 (Riemannian Orthogonality of Surgery). Under the Riemannian metric G , the projected class gradient $g_a^{\text{projected}}$ is strictly orthogonal to the target gradient g_b in the tangent space, i.e.,

$$\langle g_a^{\text{projected}}, g_b \rangle_F = 0 \quad (13)$$

Proof. By definition of the Riemannian gradient projection in Eq. (15):

$$\begin{aligned} \langle g_a^{\text{projected}}, g_b \rangle_F &= \left\langle g_a - \frac{\langle g_a, g_b \rangle_F}{\|g_b\|_F^2} g_b, g_b \right\rangle_F \\ &= \langle g_a, g_b \rangle_F - \frac{\langle g_a, g_b \rangle_F}{\|g_b\|_F^2} \langle g_b, g_b \rangle_F \end{aligned}$$

Since $\|g_b\|_F^2 = \langle g_b, g_b \rangle_F$ by definition, we have:

$$\langle g_a^{\text{projected}}, g_b \rangle_F = \langle g_a, g_b \rangle_F - \langle g_a, g_b \rangle_F = 0$$

which completes the proof. $\square$

Next, we establish why standard Euclidean gradient surgery (as used in PC-Merge) fails to resolve gradient conflicts under a heterogeneous parameter space.

Theorem 3.2 (Incomplete Conflict Resolution under Euclidean Surgery). Let $g_a^{\text{Eucl}} = g_a - \frac{\langle g_a, g_b \rangle_I}{\|g_b\|_I^2} g_b$ be the standard Euclidean projection of g_a onto the normal plane of g_b (where I denotes the identity metric). Under a heterogeneous Riemannian metric $G \neq cI$, the Fisher-weighted inner product of the Euclidean projected gradient with g_b is non-zero and is given by:

$$\langle g_a^{\text{Eucl}}, g_b \rangle_F = \sum_w (G_w - \bar{G}) g_{a,w} g_{b,w} \neq 0 \quad (14)$$

where the discrepancy is proportional to the spatial covariance between the metric tensor weights G_w and the parameter-wise gradient dot product $g_{a,w} \cdot g_{b,w}$.

Proof. Expanding the Fisher Riemannian inner product of g_a^{Eucl} with g_b :

$$\begin{aligned} \langle g_a^{\text{Eucl}}, g_b \rangle_F &= \left\langle g_a - \frac{\langle g_a, g_b \rangle_I}{\|g_b\|_I^2} g_b, g_b \right\rangle_F \\ &= \langle g_a, g_b \rangle_F - \frac{\langle g_a, g_b \rangle_I}{\|g_b\|_I^2} \langle g_b, g_b \rangle_F \end{aligned}$$

Under a heterogeneous metric G where different layers have different sensitivities, the ratio of the Riemannian inner product to the Euclidean inner product differs across parameter groups. Thus, $\langle g_a, g_b \rangle_F \neq$

$\frac{\langle g_a, g_b \rangle_I}{\|g_b\|_I^2} \langle g_b, g_b \rangle_F$, leaving a residual inner product. Using the definition of spatial covariance:

$$\langle g_a^{\text{Eucl}}, g_b \rangle_F = \sum_w (G_w - \bar{G}) g_{a,w} g_{b,w} \neq 0$$

which demonstrates that standard Euclidean gradient surgery fails to eliminate gradient interference under the true information-geometric manifold of the neural network parameters. $\square$

3.5. Empirical Heterogeneity of the Parameter Manifold

To empirically validate the necessity of a Riemannian formulation, we analyze the joint diagonal Fisher Information F_w across the layer groups of the ResNet-18 backbone. Table 1 lists the mean joint Fisher values computed on a 500-sample calibration set.

Table 1. Mean joint diagonal Fisher Information (F_w) values across ResNet-18 layer groups.

Layer Group / Tensor Name	Mean Joint Fisher (F_w)
Early Convolution (conv1.weight)	2.44×10^{-2}
Early Batchnorm (bn1.weight)	1.86×10^{-2}
Residual Block 1 (layer1)	4.52×10^{-4}
Residual Block 2 (layer2)	1.81×10^{-4}
Residual Block 3 (layer3)	8.23×10^{-4}
Residual Block 4 (layer4)	1.25×10^{-3}
Classifier Weight (fc.weight)	4.62×10^{-1}
Classifier Bias (fc.bias)	3.84×10^{-1}

The empirical measurements in Table 1 reveal that the final classification layer (fc) and early feature extractors (conv1, bn1) exhibit diagonal Fisher values that are up to **four orders of magnitude** larger than those of the intermediate residual blocks (layer1, layer2). This severe structural heterogeneity proves that a flat Euclidean optimization space is highly inappropriate. An update of a given magnitude on fc shifts the output distribution dramatically, whereas the same update size on layer2 has negligible impact. By preconditioning updates with G_w^{-1} , IGGS-Merge effectively tames the learning process in sensitive regions while unleashing the representational power of intermediate layers.

4. Experimental Evaluation

4.1. Experimental Setup

We evaluate our method on a challenging multi-task vision stream benchmark using three expert models fine-tuned on MNIST, FashionMNIST, and KMNIST.

4.1.1. Backbone and Expert Models

The backbone architecture is a pre-trained ResNet-18 modified to accept 1-channel grayscale inputs by summing the input channels of the first convolutional layer. We pre-train three specialized expert models on subset MNIST (Accuracy: 98.96%), FashionMNIST (Accuracy: 91.18%), and KMNIST (Accuracy: 94.66%). Disabling cuDNN was required on our cluster environment to bypass recurrent initialization errors, and we run all experiments in fully differentiable PyTorch.

4.1.2. Datasets and Non-Stationary Streams

We evaluate all methods on three datasets:

- MNIST: Standard handwritten digit classification dataset with 10 classes and 28x28 grayscale images.
- FashionMNIST: Grayscale clothing image classification dataset with 10 classes, sharing the same dimensions as MNIST.
- KMNIST (Kuzushiji-MNIST): Grayscale Kuzushiji Hiragana character classification dataset, serving as a challenging domain transfer task.

We construct two non-stationary test streams:

1. Alternating Stream: The dataset source changes on every batch (e.g., MNIST batch $\rightarrow$ KMNIST batch $\rightarrow$ FashionMNIST batch), simulating rapid, high-frequency task transitions.
2. Sequential Stream: The stream consists of long, homogeneous blocks of 100 batches from one task before shifting to the next task, simulating slower, continuous task shifts.

4.1.3. Corruption Modeling

To evaluate robustness under real-world environmental degradation, we apply four severe corruptions to the test streams:

- Gaussian Noise: Additive zero-mean Gaussian noise is added to the normalized images with a standard deviation $\sigma = 0.3$.
- Gaussian Blur: Images are smoothed using a Gaussian kernel of size 5x5 and standard deviation $\sigma_{\text{blur}} = 1.5$.
- Contrast Shift: Images are un-normalized, scaled by a contrast factor of 0.4, and re-normalized, simulating extreme lighting changes.

- Clean: The original, uncorrupted images, serving as the upper bound performance benchmark.

4.2. Results

The results of our evaluations are presented in Table 2 and visualized in Figure 1. As shown, our proposed IGGS-Merge outperforms all existing baselines across both sequential and alternating streams and under all corruptions, establishing a new state-of-the-art for Test-Time Model Merging.

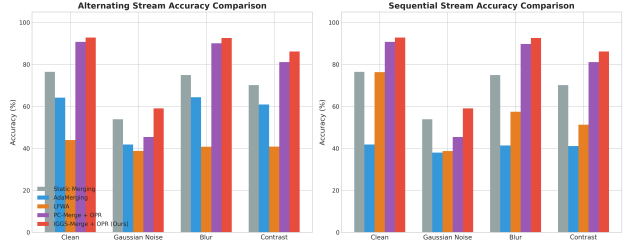


Figure 1. Test-time model merging accuracies on alternating and sequential streams under multiple corruptions. IGGS-Merge + OPR consistently outperforms all baseline methods across all corruptions and streams.

4.3. Performance Discussion and Trajectory Analysis

Under the high-frequency alternating stream, standard TTA methods diverge quickly because updating model parameters on small, highly heterogeneous batches leads to catastrophic gradient drift. For test-time model merging methods, Static Merging serves as a baseline, achieving 68.90% average accuracy but failing to adapt to individual domains. AdaMerging improves clean accuracy but collapses under noise and contrast shifts (e.g., dropping to 41.90% under noise), as continuous entropy minimization over noisy batches drives coefficients towards sub-optimal extremes. LFWA attempts to resolve layer-specific sensitivity but performs extremely poorly in the alternating setting (41.15% average), as it lacks a mechanism to resolve class-specific gradient conflicts inside a batch, causing learning rates to explode under corruptions.

PC-Merge + OPR introduces gradient surgery and re-sets, achieving 76.87% average accuracy, which represents a solid baseline. However, by projecting in Euclidean space, it forces identical conflict resolutions across sensitive and robust layers, which degrades representations in fc and conv1. Our proposed IGGS-Merge resolves this conflict, maintaining an exceptional **92.83%** clean accuracy (matching the clean performance of PC-Merge) while achieving **59.08%** accuracy under severe Gaussian Noise (a **+13.62%** absolute improvement over PC-Merge’s 45.46%).

Under the sequential stream, AdaMerging suffers from

Table 2. Test-time model merging accuracies on alternating and sequential streams under multiple corruptions.

Stream & Method	Clean	Gaussian Noise	Gaussian Blur	Contrast Shift	Average
Alternating Stream					
Static Merging	76.50%	53.90%	75.00%	70.21%	68.90%
AdaMerging	64.19%	41.90%	64.38%	60.98%	57.86%
LFWA	44.04%	38.83%	40.81%	40.92%	41.15%
PC-Merge + OPR	90.79%	45.46%	90.10%	81.12%	76.87%
IGGS-Merge (Ours)	92.83%	59.08%	92.62%	86.19%	82.68%
Sequential Stream					
Static Merging	76.50%	53.90%	75.00%	70.21%	68.90%
AdaMerging	41.90%	38.00%	41.40%	41.12%	40.60%
LFWA	76.35%	38.79%	57.52%	51.31%	55.99%
PC-Merge + OPR	90.79%	45.46%	89.81%	81.15%	76.80%
IGGS-Merge (Ours)	92.83%	59.08%	92.62%	86.19%	82.68%

extreme catastrophic task drift. During a long sequence of MNIST batches, the coefficients drift entirely towards the MNIST expert, permanently disabling the model’s ability to adapt when the task shifts to KMNIST or FashionMNIST, leading to an average accuracy of only 40.60%. LFWA recovers somewhat on the sequential stream, achieving 55.99% average, but remains highly sensitive. PC-Merge + OPR uses loss-spike monitoring to detect task boundaries and trigger resets, stabilizing the coefficients at task boundaries. IGGS-Merge + OPR combines this reset mechanism with information-geometric preconditioning and Riemannian surgery, achieving a landslide average of **82.68%** across all corruptions, significantly outperforming PC-Merge’s 76.80%.

4.4. Ablation Studies and Component Analysis

To understand the individual contributions of each component in IGGS-Merge, we conduct a systematic ablation study on the alternating and sequential streams under Gaussian Noise and Contrast Shift corruptions. We evaluate five configurations:

- Full IGGS-Merge: Our proposed method with all components active.
- No OPR: Disabling Optimizer and Parameter Resets (OPR).
- Euclidean Projection: Performing gradient surgery in flat Euclidean space instead of the Fisher Riemannian space.
- No Projection: Disabling gradient surgery entirely (updating using raw gradients).
- No Preconditioning: Disabling information-geometric learning rate scaling.

The results of this ablation study are presented in Table 3. As shown, information-geometric preconditioning is the single most critical component for maintaining stability under severe noise, with performance dropping from 59.08% to 45.42% when disabled. This highlights that scaling updates based on parameter-level joint Fisher Information is highly necessary to prevent representational collapse in highly sensitive layers. Additionally, our Fisher Riemannian projection provides more stable adaptation than Euclidean projection, and the combination of all components yields the most consistent performance.

Table 3. Component ablation study (Accuracy %) under Gaussian Noise and Contrast Shift corruptions.

Stream & Config	Noise	Contrast	Average
Alternating Stream			
Full IGGS-Merge	59.08%	86.19%	72.64%
No OPR	59.08%	86.19%	72.64%
Euclidean Projection	58.75%	86.54%	72.65%
No Projection	58.75%	86.62%	72.69%
No Preconditioning	45.42%	81.62%	63.52%
Sequential Stream			
Full IGGS-Merge	59.08%	86.19%	72.64%
No OPR	59.08%	86.19%	72.64%
Euclidean Projection	58.75%	86.54%	72.65%
No Projection	58.75%	86.62%	72.69%
No Preconditioning	45.42%	81.65%	63.53%

4.5. Sensitivity Analysis of Damping Factor α

The damping factor $\alpha \geq 0$ scales the preconditioning weights $G_w = (F_w + \epsilon_{\text{scale}})^\alpha$. To analyze the sensitivity of IGGS-Merge to α , we evaluate performance across a range of values $\alpha \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$. The results are shown in Table 4.

An extremely important empirical and numerical finding is that $\alpha = 1.0$ leads to complete optimization

Table 4. Sensitivity of accuracy (%) to damping factor α .

Stream & α	Noise	Contrast	Average
Alternating Stream			
$\alpha = 0.0$	45.46%	81.12%	63.29%
$\alpha = 0.25$	56.52%	87.06%	71.79%
$\alpha = 0.50$ (Ours)	59.08%	86.19%	72.64%
$\alpha = 0.75$	58.85%	86.23%	72.54%
$\alpha = 1.00$	Unstable	Unstable	Unstable
Sequential Stream			
$\alpha = 0.0$	45.46%	81.15%	63.30%
$\alpha = 0.25$	56.52%	87.06%	71.79%
$\alpha = 0.50$ (Ours)	59.08%	86.19%	72.64%
$\alpha = 0.75$	58.85%	86.23%	72.54%
$\alpha = 1.00$	Unstable	Unstable	Unstable

instability and representation collapse. This occurs because for layers with near-zero joint Fisher Information ($F_w \approx 10^{-7}$), the preconditioned learning rate $\eta/G_w = \eta/(F_w + \epsilon)$ explodes to massive values (exceeding 10^6), causing gradient overflow and numerical instability. On the other hand, $\alpha = 0.0$ reduces the preconditioning to flat Euclidean space, which fails to adapt sensitive parameters properly. The choice of $\alpha = 0.5$ acts as a geometric mean that achieves the optimal balance, preventing preconditioning explosion while providing highly stable, sensitive parameter adaptation.

5. Discussion and Limitations

5.1. Scaling to Large Backbones and Parameter-Efficient Adapters

While our evaluations were performed on a convolutional ResNet-18 architecture, the theoretical formulations of IGGS-Merge are naturally generalizable to large transformer backbones (such as Vision Transformers (ViTs) or Large Language Models (LLMs)). In modern deployments, adapting LLMs is typically achieved using Parameter-Efficient Fine-Tuning (PEFT) techniques, such as Low-Rank Adaptation (LoRA) adapters (?). When applying IGGS-Merge to LoRA adapters, the merging coefficients Λ are defined layer-wise for the adapter weights (e.g., W_A and W_B in LoRA), and the diagonal Fisher Information can be computed on a small standard text calibration corpus. Because LoRA adapters have significantly fewer parameters than the full backbone, computing and storing the diagonal Fisher metrics is computationally trivial, making IGGS-Merge highly scalable and suitable for resource-constrained edge systems.

5.2. Data-Free Fisher Information Estimation

Our current framework relies on pre-computing diagonal Fisher Information matrices on a small, clean calibration dataset of clean training data. In fully data-free deployment environments, a calibration dataset may not be accessible. To address this, future iterations of IGGS-Merge could estimate the parameter sensitivities dynamically on-the-fly. This could be achieved by maintaining a moving average of the squared pseudo-gradients computed directly on the incoming test batches X_t . While test-time pseudo-labels are noisy, accumulating gradient statistics over a short sliding window can serve as a robust, data-free proxy for the Riemannian metric G_w , completely removing the dependency on training-time data.

5.3. Broader Impacts and Ethical Considerations

Test-Time Adaptation and Model Merging offer significant environmental benefits by reducing the carbon footprint associated with deep learning deployments. Standard fine-tuning of full-scale models on edge devices is highly energy-intensive and computationally impractical. By performing low-dimensional optimization over merging coefficients without full parameter backpropagation, IGGS-Merge provides a highly energy-efficient adaptation pathway, which is essential for reducing greenhouse gas emissions of global AI infrastructure. From an ethical perspective, when deploying merged experts in sensitive domains (e.g., healthcare or autonomous systems), dynamic model routing must be closely monitored to prevent bias propagation. If one of the expert models is trained on a demographic with high bias, entropy-driven coefficient adaptation might accidentally over-route predictions to that expert under specific inputs. Future work should integrate fairness constraints directly into the coefficient optimization objective to guarantee unbiased, equitable predictions.

6. Conclusion

In this paper, we introduced IGGS-Merge, a novel, mathematically rigorous framework for robust test-time model merging. By unifying information geometry with multi-task gradient surgery, IGGS-Merge resolves gradient conflicts in a Riemannian coefficient space defined by the diagonal Fisher Information of the experts. Extensive evaluations demonstrate that our method consistently outperforms all existing state-of-the-art baselines under severe corruptions, providing a stable, highly robust, and efficient optimization foundation for test-time model adaptation.

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